

Week 1 – Building Your Quantum Development Workstation

Objectives

- Understand what quantum computing is and is not.
- Install Python 3.13 (or latest stable 3.x).
- Install Visual Studio Code.
- Install Git.
- Create a GitHub repository named quantum-learning.
- Create a Python virtual environment.
- Install Qiskit, JupyterLab, Matplotlib, and Qiskit Aer.
- Verify the environment with a simple Python import test.

Tasks

1. Create folder: ~/Development/quantum-learning
2. Initialize a Git repository and push to GitHub.
3. Create a virtual environment named .venv
4. Install: qiskit qiskit-aer jupyterlab matplotlib pylatexenc
5. Open the folder in VS Code and select the virtual environment.
6. Create verify.py that imports qiskit and prints its version.
7. Run verify.py successfully.

Minimum Completion: All software installed, repository created, virtual environment working, verify.py executes successfully.